

# Standardization of commercial catch data from multiple gears in mixed fisheries accounting for preferential sampling, catchability, and fishing effort

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## ABSTRACT

Commercial fisheries constitute a valuable source of high-resolution information that can aid in assessing stocks and establishing management schemes. Especially, multi-gear and multi-species fisheries can provide fine-scale insights in space and time with regards to the patterns in species distribution and abundance as well as to the comparative behavior of the fishing gears deployed. In this work we propose a Generalized Additive Modeling framework to standardize catch data collected through observer monitoring programs using a 2018–2021 dataset from the eastern Ionian (Mediterranean Sea, FAO GFCM GSA20) as a case study of data-poor mixed fisheries. Our framework extends the standardization procedures by accounting for preferential sampling, integrating effort from multiple gears and jointly modeling species. We show that such an integration leads to more robust estimations of abundance for both target and by-catch species as well as decreases inference uncertainty. Regarding single stocks, the identification of the independent effect of factors (e.g. spatial, temporal, fishing effort, gear, skipper effect) can aid in monitoring and management decisions; furthermore, an objective index of abundance is estimated that can be used to infer inter-annual trends from more extended time-series useful for stock assessments. Using standardized catch values, we have generated seasonal maps of species distribution and multiple-species persistence hotspots that are useful for designing spatiotemporal management restrictions and also informative of species ecology. We also address the effect of the technical (selectivity) and behavioral aspects of the fishing gears to inform gear-based management. Finally, we demonstrate how this broad inferential process can be condensed to form species assemblages (based on their shared responses on drivers of catch and abundance) as well as fishing gear assemblages (based on their catch profiles and the apparent heterogeneity between vessels deploying common gears) that can act as units of reference for management. Apart from an objective estimation of stock abundance in time and space, our standardization framework illustrates how ecological, technical and behavioral aspects of mixed fisheries can be collectively evaluated to inform stock assessment and management.

## 1. Introduction

Fisheries stock assessments and management require input from experimental surveys and commercial fisheries catch composition data. The input from these data can range from temporal (inter-annual) trends of stock abundance (Maunder and Punt, 2013; Campbell, 2015) to species distributional patterns (Pennino et al., 2019; Alglave et al., 2022, Paradinas et al., 2023) that provide insight to the stock status as well as aid in optimizing management plans. The catch (C) of a fishing

operation is the product of effort (E), catchability (q) and population abundance (N) for a given area and time,  $C = E \times q \times N$  (Maunder et al., 2006). Catch per unit of effort (CPUE) can be used as a proxy for the abundance of a stock after controlling for catchability which involves ecological (species interaction with environmental factors), technical (fishing gear characteristics) and behavioral (subjective choices adopted by professional fishermen) components. The last two components are mostly relevant when analyzing commercial fisheries data due to the inherent bias that accompanies the selectivity of the gears employed by

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fisher decisions (Maunder and Punt, 2004; Hoyle et al., 2024). Catch standardization is performed through regression modeling techniques that aim to control for the effect of a variety of environmental and catchability covariates implemented both in frequentist and bayesian frameworks. Fishing effort is usually implemented either as an offset covariate since it refers to the rate of the response variable (catch) or as a usual covariate whose coefficient is to be estimated if not a constant one-to-one relationship between catch and effort is assumed. Hoyle et al. (2024) provide a thorough review for the methods and considerations needed when standardizing catch data.

Mixed fisheries are fisheries that exploit more than one stock simultaneously (Ulrich et al., 2017; Sun et al., 2023). This inherent characteristic renders them heterogeneous, as abundance and catch fluctuations are apparent for each stock participating in the catches. Apart from differences in abundance, the various species may have diverse behavioural patterns, varying degrees of vulnerability and catchability (Samy-Kamal et al., 2015); these further complicate the collection of data sufficient for stock assessments and also impede the design of management schemes that are suitable for all species involved (Dolder et al., 2018). The situation faced by scientists and managers may be even more complicated if a variety of fishing gears target or catch some or all of these stocks as the different gears may have technical interactions and partial overlaps in time and space (e.g. Pappa et al., 2017).

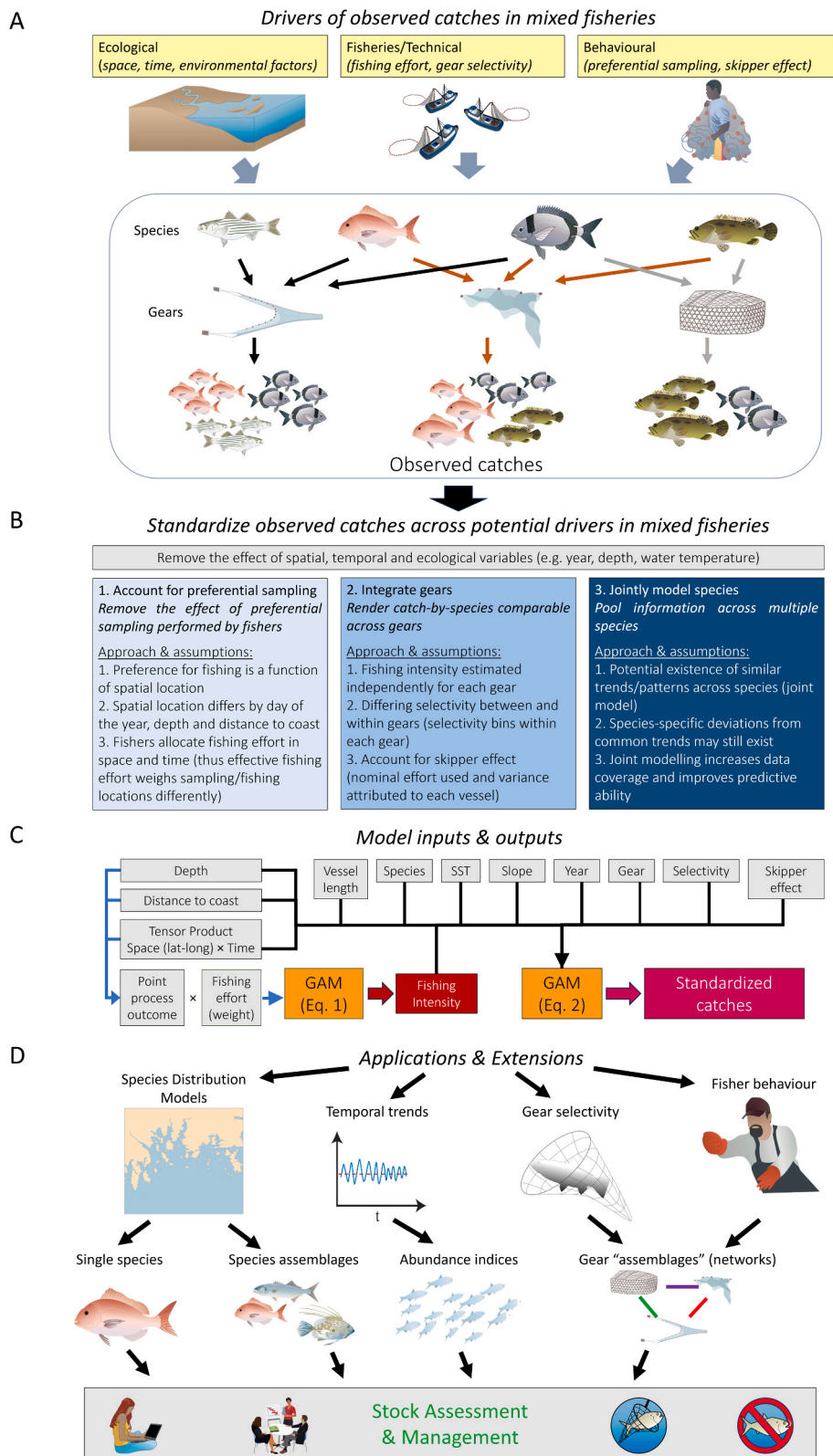
Typical examples of fisheries of the mixed character mentioned above are most fisheries in the Mediterranean Sea and in southern Europe. Most fisheries of this area are multi-species, i.e. the catch of a specific gear is composed of a variety of species, and multi-gear, i.e. the same species appears in the catch of (and is often targeted by) various fishing gears (Tzanatos et al., 2013). The multi-gear and multi-species character of Mediterranean fisheries have inherent heterogeneity and thus uncertainty attributes, whose effects are not additive, but rather multiplicative to each other, since more-than-one gears may target the same more-than-one species. With the exception of a handful of gears that target and catch specific stocks with a few bycatch species, like lines for bluefin tuna and longlines for swordfish, most fishing gears that operate in the Mediterranean Sea (industrial ones like bottom trawls and purse seines and artisanal ones like gillnets, trammel nets and set longlines) not only catch various species, but also often show overlap in catching some of them.

In the multi-gear/multi-species fisheries context, we observe variety both in the fishing gears and in the captured species. On one hand, since multiple different fishing gears operate on a common stock, there is potentially a variety of sources in assessing the status of a stock as each gear contains potential information on a species ecology (e.g., Bourdaud et al., 2017) since each fishing gear obtains a different snapshot of the population with regards to space and time. On the other hand, since a variety of species appear in the catch of a fishing gear (whether targeted or not), we can also assess the efficacy and behavior of a fishing gear by examining its catches across many species. The captured species may or may not share abundance patterns, but, nonetheless, the total catch profile of a gear can approximate its “technical profile”, depicting the interaction of the gear with these (multiple) species examined within a management context. In species distribution ecological models, methods have been proposed to incorporate information from multiple sources that range from the pooling of data sources (Cheng et al., 2023) to model averaging (Dormann et al., 2018) and more sophisticated integration techniques (Fletcher et al., 2019). Specifically in the context of fisheries, various works have proposed the increase in the accuracy of the assessment evaluated after integrating commercial and survey fisheries data (Simmonds et al., 2020; Martin Gonzalez et al., 2021; Paradinas et al., 2023; Alglave et al., 2023). Furthermore, jointly modeling data for many species available (Joint Species Distribution Models, JSDMs), with appropriate correlation structures among the species in the modeling procedure, has been established as a way to share information between

species and overcome species-specific data scarcity (Warton et al., 2015; Thorson et al., 2017; Roberts et al., 2022).

The idea of integrating a variety of sources in order to make inferences about the stocks seems promising for the cases where information is obtained from different fishing gears and for multiple species as, for example, in observer monitoring programs of mixed fisheries. Incorporating catch information available for all gears in a model is a way to make better estimates of catch at a unified joint spatio-temporal grid (Kinoshita et al., 2019) rather than combining catch estimates from separate models of each fishing gear (Fletcher et al., 2019). A main obstacle to overcome in such an integration is the lack of a common metric for fishing effort. Fishing effort is often expressed nominally using, e.g., capital (vessel characteristics) or a measure of fishing intensity relevant to the operation of the employed gear such as net length, number of hooks or trawling duration (McCluskey and Lewison, 2008; Aljafary et al., 2019). Thus, assessing the catch rate, in the form of CPUE, on the level of a fishing trip, differs and is not comparable among gears since the quantification of fishing effort is gear-specific (Padilla and Trinidad, 1995) and a simple inter-gear pooling of data is not considered effective (Gibson-Reinemer et al., 2017; Fletcher et al., 2019). The fishing effort of each fishing gear can be thought of as the last of the three main hierarchical levels where sampling design in commercial fisheries is actually carried out. For ease of reference, we can arbitrarily separate the sampling design decisions that need to be taken, in three levels: (i) the base of the sampling process (see Cotter and Pilling, 2007, for a thorough breakdown), (ii) the deployment of survey scientists (Benoît and Allard, 2009; Faunce and Barbeaux, 2011; Duarte and Cadrin, 2024) and (iii) the actual sampling of the stock through fishing. Since fishing generally aims to maximize catches, fishing effort is not randomly allocated in space and time, but is rather biased towards higher stock abundance (Hilborn and Walters, 1987; Maunder and Punt, 2004). This type of bias is termed preferential sampling (Diggle et al., 2010) and has been shown to influence the results when analyzing opportunistic data (Watson, 2021). Preferential sampling has only recently been taken into account, both in the broader species distribution modeling scene (Conn et al., 2017; Watson, 2021) and in fisheries in particular (Pennino et al., 2019; Alglave et al., 2022; Ducharme-Barth et al., 2022). Preferential sampling has been shown to lead to biased estimates when it is not accounted for (Diggle et al., 2010; Conn et al., 2017).

In the current work we aim to provide a framework where intricacies of mixed fisheries can be dissected and delivered to practical assessment and management advice (Fig. 1). We focus on how commercial catch data, collected through observer monitoring programs, can be utilized in a reliable way in order to examine the behavior of fish stocks and the behavior of fishing gears so that appropriate management advice can be developed either on the stock or the gear level, respectively. For this task, we present a two-fold enhancement of inference from commercial fisheries catch data: (i) a modeling-enhancement that allows inclusion of all available data and (ii) an inference-enhancement that organizes outputs from a multi-component model in order to be interpretable. Specifically, on the one hand, we develop a standardization procedure of commercial catch and effort data that, along with common drivers of abundance and actual catch, integrates information among multiple fishing gears and multiple captured species, while addressing the effect of preferential sampling as discerned through the specific fishing effort of each gear. On the other hand, these standardized data can then be used to a) provide objective (single or multiple) species estimations of abundance, b) quantify species distributional patterns, c) determine groups of species with common dynamics across environmental or fisheries-related attributes and d) evaluate the catchabilities of the fishing gears in conjunction with behavioral aspects of fishing (“skipper” effect). These elements can be incorporated in the assessment of the exploited stocks and the management of the fishing gears in space and time.



All images in this figure are courtesy of the Integration and Application Network, University of Maryland Center for Environmental Science ([ian.umces.edu/symbols/](http://ian.umces.edu/symbols/))

**Fig. 1.** Conceptual diagram of the rationale, approach and applications of the catch standardization developed in this work. A. Examples of the multitude of drivers (ecological, technical, fisher behaviour) shaping the observed catches of mixed fisheries. B. Outline of the “typical” standardization process (across spatial, temporal, ecological variables) and the main axes developed here (preferential sampling account, gear integration and joint modelling of species) along with the approach and main assumptions. C. Modelling flowchart from covariates to standardized catches: thin arrows pointing to model boxes indicate model inputs, while outgoing thick arrows indicate outputs. D. Schematic presentation of the applications and extensions in the use of standardized catches to inform stock assessments and management scheme design.

## 2. Materials and methods

### 2.1. Data collection and analysis framework

Sampling was carried out onboard commercial fishing vessels during 2018–2021 (also including the first trimester of 2022) in the eastern Ionian Sea, a Geographic Sub-Area (GSA 20) as recognized from the General Fisheries Commission for the Mediterranean (FAO GFCM), under the auspices of the European Union Data Collection Framework (EC 199/2008, DCF). The collected data were referenced at the haul level, i.e. a specific fishing operation during a vessel trip. Each haul was defined by a specific gear type, the haul spatial and temporal attributes as well as its catch composition by species. Table 1 shows the total number of trips, for each fishing gear, surveyed during the sampling period as well as the estimated number of total trips in GSA 20 from the DCF report.

#### 2.1.1. Area and fisheries characteristics

The GSA 20 is divided in three spatial divisions (North, Central and South Ionian Sea) that allows for a more robust sampling design, under the DCF, especially concerning the small-scale fisheries as the small-scale fishing vessels generally operate close to their home port in nearby fishing grounds (e.g. Tzanatos et al., 2006). The main gears operating in the area are gillnets, trammel nets, bottom longlines, bottom trawls and purse seines. Each gear is subject to different management measures that affect its operations both spatially and temporally. This spatial stratification of the sampling scheme followed by the DCF leads to “spatial gaps” of fishing operations connecting the three divisions in space (Fig. 2). This results in the data being spaced in three distinct patches and, hence, a spatial fine-scale analysis for the entirety of GSA 20 was not deemed appropriate. So, in our current work, we split the datasets of each division and analyzed each division separately. This offers the ability to utilize a finer spatial resolution rather than, for example, clustering fishing operations to larger spatial domains.

Hauls, for each fishing gear, that were considered as outliers due to either their significantly larger catches compared to the rest (difference in ~2 orders of magnitude, i.e. positioned in the 98<sup>th</sup> percentile of catches and beyond) or their spatial remoteness from the centroid of each division were discarded from analysis (Suppl. Figure S1). In addition, we selected the subset of species that occurred most frequently in the fishing operations of each division in GSA 20 (regardless of gear

type). Specifically, for each division, we computed the percentage of hauls that each species appears in and selected those species whose frequency is above the 90<sup>th</sup> percentile of all species frequencies. Since our aim was to include species with a wide data coverage we computed frequencies of appearance in the total number of hauls irrespective of fishing gear rather than on a gear level (since the latter would lead to missing important commercial species if they did not appear consistently in the catches of all fishing gears equally). Table 2 presents the number of hauls by gear for each GSA 20 division as well as the species selected for the analyses along with their respective frequency in the hauls of each fishing gear.

#### 2.1.2. Preferential sampling and data integration

In the present work we accounted for the preferential allocation of fishing effort and gear deployment which is potentially correlated with fish abundance. Preferential sampling can be accounted for by using a marked point process pattern to model the fishing intensity, as a unitless outcome variable, along spatial and/or temporal components, as well as any other confounding factors (Diggle et al., 2010; Youngman and Economou, 2017). Since the observed variable of interest (here, the catch of a haul, the mark of a spatial point pattern) dictates the fishing intensity, it is essential to model them together. This approach has been adopted under the Bayesian framework using joint likelihood models (Pennino et al., 2019; Vedensky et al., 2023). Here, we used a two-step approach: firstly, we modeled fishing intensity and then, used the predicted intensity as a control covariate in modeling catch by species (Si et al., 2015; Vedensky et al., 2023). By modeling the fishing effort of each fishing gear as a point process and projecting the fishing intensity as a rate, we directly quantified the rates of fishing intensity as comparable among the different gears and used this “latent” quantification of effort in our integrated catch standardization model. Pooling information across species was performed by utilizing a hierarchical mixed model framework where the species-specific responses are considered as deviations from an average response. Integrating data from multiple species enabled us to disentangle similar covariate effects among species by sharing inter-specific information and obtain better predictive performance overall (Jamil et al., 2013; Fithian et al., 2015; Roberts et al., 2022).

#### 2.1.3. Selectivity bias

Management schemes applied to the different fishing gears impose technical constraints (e.g. hook/mesh size) which result in selectivity constraints when sampling stocks. In the current work we opted to analyze the fishing gear selectivity as a categorical variable (and not as a continuous one, albeit recorded as such during observer monitoring programs), since, in métier classifications, gear selectivity covers broad ranges of mesh and hook sizes (e.g. Tzanatos et al., 2006). We discretized the selectivity metric for each gear into bins (ranges of numerical value into which data were sorted) that were deemed more appropriate both biologically (rugged selectivity thresholds commonly followed by professional fishers when targeting species) and statistically (avoid high heterogeneity in the sizes among bins). The selectivity bins (right-closed intervals) we used were: (i) for gillnets/trammel nets:  $<=20$ ,  $(20, 26]$ ,  $(26, 32]$ ,  $(32, 36]$ ,  $(36, 44]$ ,  $>44$  mm of mesh size, (ii) for longlines:  $<=8$ ,  $(8, 11]$ ,  $(11, 14]$ ,  $>14$  of hook size (a smaller manufacture number is a physically larger hook). We used a single bin for (iii) bottom trawlers and (iv) purse seines since these gears are legally restricted to a minimum mesh size commonly used (45 mm for trawlers, 18 mm for purse seines).

#### 2.1.4. Other covariates influencing observed catch and abundance

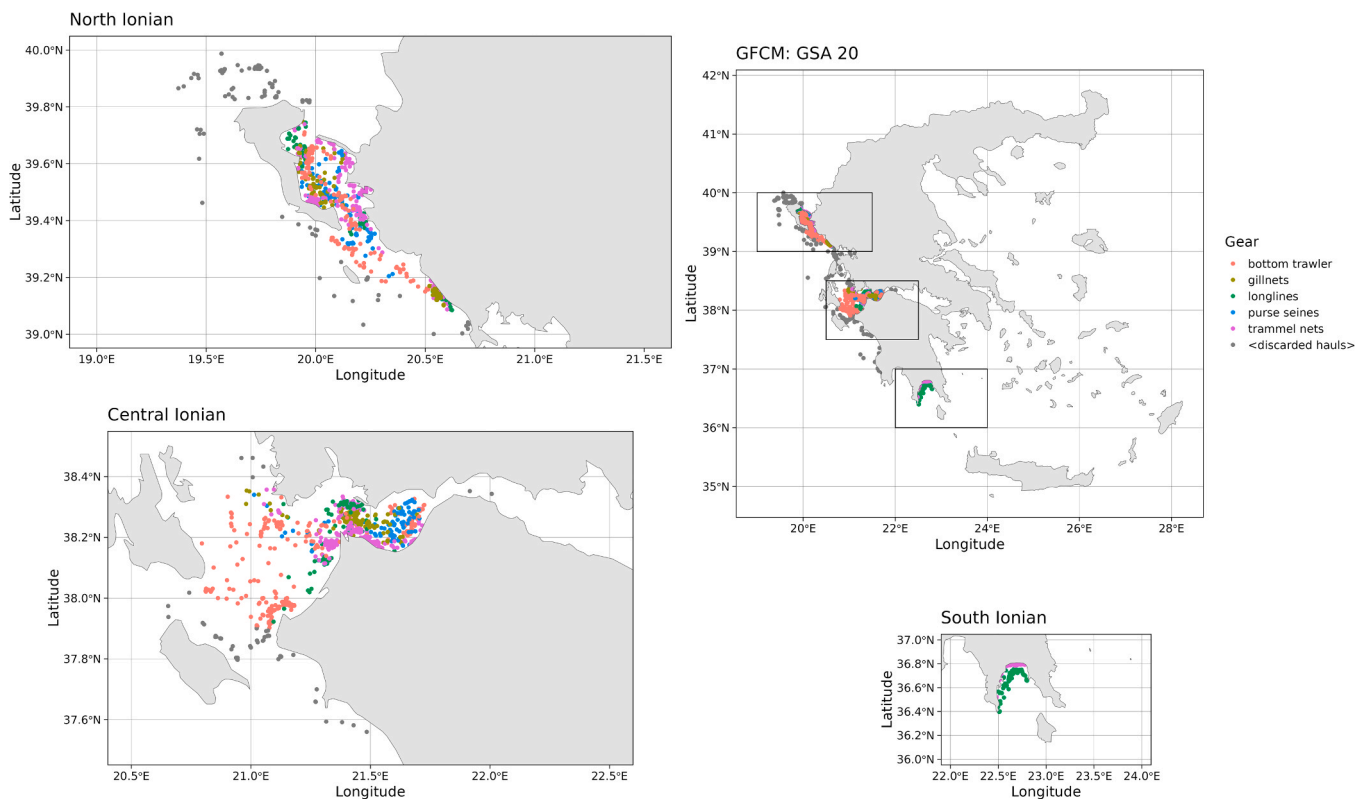
Relevant recorded haul-referenced data during the observer monitoring programs include information on the nominal technical characteristics of a vessel, the fishing gear deployed along with its selectivity metric, the depth and the spatio-temporal stamp of the haul. Data on other potential drivers of fish distribution, abundance and capturing by

**Table 1**

Number of sampled hauls, by fishing gear and year, during the observer surveys under the Greek Data Collection Framework. The total estimated number of hauls are presented in parentheses and are based on official DCF reporting.

|                  |      | Region       |                |              |
|------------------|------|--------------|----------------|--------------|
| Gear             | Year | North Ionian | Central Ionian | South Ionian |
| Gillnets         | 2018 | 36 (37864)   | 29 (27384)     | - (6288)     |
|                  | 2019 | 27 (34422)   | 25 (32511)     | - (8456)     |
|                  | 2020 | 30 (30989)   | 32 (23002)     | - (8699)     |
|                  | 2021 | 33 (30719)   | 24 (17404)     | - (11913)    |
| Trammel nets     | 2018 | 122 (95988)  | 62 (73234)     | 33 (29071)   |
|                  | 2019 | 102 (79726)  | 39 (61282)     | 13 (27625)   |
|                  | 2020 | 131 (71585)  | 52 (59219)     | 5 (25225)    |
|                  | 2021 | 143 (60590)  | 62 (74647)     | 12 (28542)   |
| Bottom longlines | 2018 | 21 (56218)   | 21 (54610)     | 22 (8904)    |
|                  | 2019 | 31 (51558)   | 25 (51837)     | 24 (9529)    |
|                  | 2020 | 24 (53032)   | 16 (52912)     | 23 (7312)    |
|                  | 2021 | 19 (64920)   | 18 (50812)     | 23 (8475)    |
| Otter trawls     | 2018 | 49 (1682)    | 57 (3693)      | - (352)      |
|                  | 2019 | 42 (1593)    | 53 (3527)      | - (306)      |
|                  | 2020 | 26 (1460)    | 64 (3869)      | - (315)      |
|                  | 2021 | 31 (1635)    | 63 (3728)      | - (320)      |
| Purse seines     | 2018 | 36 (2023)    | 28 (2785)      | - (60)       |
|                  | 2019 | 44 (1846)    | 28 (4305)      | - (121)      |
|                  | 2020 | 25 (1237)    | 17 (2688)      | - (68)       |
|                  | 2021 | 17 (1674)    | 9 (2357)       | - (94)       |





**Fig. 2.** Map of the eastern Ionian Sea (FAO GFCM GSA 20) divided in three subdivisions for sampling purposes under the European Union Data Collection Framework (DCF, EC 199/2008). Individual fishing trips (hauls) spanning the period 2018–2021 are depicted with different colors by gear. Hauls that were considered as outliers and were discarded from analysis due to their significantly larger catches compared to the rest or their spatial remoteness from the centroid of each division are also shown (see, also, Suppl. Fig. S1).

fisheries were retrieved from online databases. Specifically, depth (used for predictive purposes in un-fished areas) and bathymetric slope were retrieved from MARSPEC (Sbrocco and Barber, 2013), sea surface temperature (SST) was retrieved from NOAA (Huang et al., 2021) and coastline data (to estimate the distance of a haul to its nearest coast) were retrieved from Natural Earth databases (<https://www.naturalearthdata.com/>; Massicotte and South, 2023).

Nominal vessel technical characteristics available in our data were the vessel length, horsepower and gross tonnage. Since in our dataset, all three metrics were highly correlated, we used vessel overall length as a proxy for the nominal effort in our analysis, since it is easily obtainable and verified during observer surveys. An alternative could be to either use more than one metric (as a univariate or interaction effect term) or to perform a Principal Component Analysis and retain the component that better describes the nominal effort exerted.

## 2.2. Statistical analysis

As mentioned previously, we analyzed each GSA 20 division separately. For each division, we first examined the fishing intensity of each fishing gear and then used fishing intensity as a control covariate in standardizing the catches in each division. We developed a framework using Generalized Additive Models (GAMs – Hastie and Tibshirani, 1986; Wood, 2011) since they are commonly used in ecology (Williamson et al., 2022) and easily applicable in fisheries science in particular (see, e.g. Maunder and Punt, 2004; Rooper et al., 2016; Grüss et al., 2017; Stock et al., 2020; Zhou et al., 2019; Martin Gonzalez et al., 2021; Cheng et al., 2023; Hoyle et al., 2024). All data preparation and analyses were performed in R Core Team (2023); we used the “mgcv” package for GAMs (Wood, 2011), the “ggplot2” package for graphical

presentations (Wickham, 2016) and the “sfdep” package for hotspot analysis (Parry and Locke, 2024).

### 2.2.1. Fishing intensity

**2.2.1.1. Point process model.** To model fishing intensity we used a Generalized Additive point process model as described in Youngman and Economou (2017). Our focus here was to disentangle the spatial and temporal clustering of fishing trips and effort. A fishing trip is ascribed a spatial and temporal bias as well as an amount of effective fishing effort allocated by fishers. For this reason, we modeled fishing intensity using a weighted point process model where the weights correspond to the effective fishing effort of a haul. We modeled the fishing intensity for each gear separately, since the unit of effective effort differs between gears (i.e., a unit of net length, a unit of hook count and a unit of trawling haul duration do not have a straightforward or easily obtained relative catchability relationship). Fishing intensity expresses the rate of occurrence (on the log scale) of a haul on the spatial and temporal grid scaled by the effective effort as defined by each gear, which, in contrast to effective fishing effort, can be comparable among gears.

In simple notation the model we used for the fishing intensity was of the form (Eq. 1):

$$\log \lambda(s, t) = w(s, t) * \{V(s, t) + \text{depth}(s) + \text{distance} - \text{to} - \text{coast}(s)\} \quad (1)$$

where:  $\lambda(s, t)$  is the intensity of the fishing in space  $s$  (longitude  $x$  and latitude  $y$ ) and time  $t$  (day of year) of the spatio-temporal  $V(s, t)$  field.

$w(s, t)$  is the weight in space  $s$  and time  $t$  as defined by the effective fishing effort deployed in each haul.

**Table 2**

The frequency of the selected species, in our analysis, by fishing gear and division GSA 20 (North, Central and South Ionian). The total number of sample hauls for each fishing gear is given below the gear. GNS: gillnets, GTR: trammel nets, LLS: bottom longlines, OTB: otter trawls, PSN: purse seines.

| Species                          | North Ionian |      |      |      |      | Central Ionian |      |      |      |      | South Ionian |      |
|----------------------------------|--------------|------|------|------|------|----------------|------|------|------|------|--------------|------|
|                                  | GNS          | GTR  | LLS  | OTB  | PSN  | GNS            | GTR  | LLS  | OTB  | PSN  | GTR          | LLS  |
| <i>Boops boops</i>               | 127          | 498  | 95   | 148  | 122  | 110            | 215  | 80   | 237  | 82   | 63           | 92   |
| <i>Citharus linguatula</i>       | 49 %         | 14 % | 12 % | 99 % | 74 % | 25 %           | 15 % | 8 %  | 87 % | 60 % | -            | -    |
| <i>Conger conger</i>             | 36 %         | 15 % | -    | 80 % | 2 %  | 90 %           | 27 % | -    | 72 % | -    | -            | -    |
| <i>Diplodus annularis</i>        | -            | -    | -    | -    | -    | -              | -    | -    | -    | -    | -            | 67 % |
| <i>Diplodus vulgaris</i>         | 51 %         | 37 % | 6 %  | 66 % | 17 % | 12 %           | 42 % | 22 % | 40 % | 2 %  | 81 %         | -    |
| <i>Engraulis encrasicolus</i>    | 9 %          | 27 % | 23 % | 11 % | 7 %  | -              | -    | -    | -    | -    | -            | -    |
| <i>Galeus melastomus</i>         | 2 %          | 2 %  | -    | 62 % | 59 % | -              | -    | -    | -    | -    | -            | -    |
| <i>Helicolenus dactylopterus</i> | -            | -    | -    | -    | -    | -              | -    | -    | -    | -    | -            | 65 % |
| <i>Illex coindetii</i>           | -            | -    | -    | -    | -    | -              | -    | -    | -    | -    | -            | 65 % |
| <i>Lepidotrigla cavillone</i>    | -            | -    | -    | -    | -    | 36 %           | -    | -    | 66 % | 40 % | -            | -    |
| <i>Loligo vulgaris</i>           | 15 %         | 7 %  | -    | 84 % | 2 %  | 13 %           | 2 %  | -    | 84 % | -    | -            | -    |
| <i>Lophius budegassa</i>         | 2 %          | 8 %  | -    | 89 % | 38 % | 6 %            | 9 %  | -    | 75 % | 65 % | -            | -    |
| <i>Merluccius merluccius</i>     | -            | -    | -    | -    | -    | 73 %           | 6 %  | 1 %  | 57 % | 1 %  | -            | -    |
| <i>Mullus barbatus</i>           | 46 %         | 11 % | 13 % | 93 % | 4 %  | 97 %           | 17 % | 21 % | 89 % | 40 % | 5 %          | 86 % |
| <i>Mullus surmuletus</i>         | 69 %         | 23 % | -    | 99 % | 15 % | 41 %           | 43 % | 1 %  | 84 % | -    | 79 %         | -    |
| <i>Octopus vulgaris</i>          | 39 %         | 24 % | 3 %  | 48 % | 12 % | 21 %           | 43 % | -    | 54 % | -    | 54 %         | -    |
| <i>Pagellus acarne</i>           | 2 %          | 12 % | 6 %  | 80 % | 4 %  | 2 %            | 30 % | 4 %  | 72 % | -    | -            | -    |
| <i>Pagellus erythrinus</i>       | 35 %         | 18 % | 8 %  | 71 % | 2 %  | -              | -    | -    | -    | -    | -            | -    |
| <i>Parapenaeus longirostris</i>  | 73 %         | 29 % | 27 % | 99 % | 13 % | 53 %           | 52 % | 30 % | 83 % | -    | 43 %         | 2 %  |
| <i>Phycis blennoides</i>         | -            | -    | -    | 84 % | -    | 22 %           | 1 %  | -    | 63 % | -    | -            | -    |
| <i>Sardina pilchardus</i>        | -            | -    | -    | -    | -    | -              | -    | -    | -    | -    | -            | 75 % |
| <i>Scomber colias</i>            | 3 %          | 2 %  | -    | 80 % | 75 % | 12 %           | 1 %  | -    | 40 % | 98 % | -            | -    |
| <i>Sepia officinalis</i>         | 24 %         | 5 %  | -    | 23 % | 61 % | 30 %           | 4 %  | 1 %  | 34 % | 67 % | -            | -    |
| <i>Serranus cabrilla</i>         | 3 %          | 58 % | -    | 92 % | 7 %  | 4 %            | 51 % | -    | 67 % | 1 %  | 54 %         | -    |
| <i>Solea solea</i>               | -            | -    | -    | -    | -    | -              | -    | -    | -    | -    | 49 %         | 3 %  |
| <i>Sparus aurata</i>             | 19 %         | 21 % | -    | 52 % | 5 %  | -              | -    | -    | -    | -    | -            | -    |
| <i>Sphyrna sphyraena</i>         | 3 %          | 21 % | 26 % | 84 % | 16 % | 31 %           | 49 % | 56 % | 70 % | 18 % | -            | -    |
| <i>Spicara flexuosa</i>          | 27 %         | 8 %  | -    | 41 % | 45 % | -              | -    | -    | -    | -    | 65 %         | -    |
| <i>Squilla mantis</i>            | 26 %         | 10 % | 1 %  | 84 % | 5 %  | 8 %            | 10 % | 5 %  | 73 % | 7 %  | 57 %         | -    |
| <i>Trachurus mediterraneus</i>   | 38 %         | 24 % | -    | 85 % | 1 %  | 69 %           | 18 % | -    | 54 % | -    | -            | -    |
| <i>Trachurus trachurus</i>       | 28 %         | 7 %  | 2 %  | 83 % | 57 % | 29 %           | 14 % | 9 %  | 79 % | 77 % | -            | -    |
| <i>Trisopterus minutus</i>       | -            | -    | -    | -    | -    | 54 %           | 2 %  | 2 %  | 63 % | 26 % | -            | -    |
| <i>Uranoscopus scaber</i>        | -            | -    | -    | -    | -    | 50 %           | 3 %  | 5 %  | 50 % | 1 %  | -            | -    |
| <i>Zeus faber</i>                | 19 %         | 18 % | -    | 81 % | 4 %  | 53 %           | 27 % | 1 %  | 66 % | -    | -            | -    |
|                                  | -            | -    | -    | -    | -    | 7 %            | 12 % | -    | 68 % | -    | -            | -    |

$depth(s)$  is the depth of the location  $s$ .

$distance\text{-}to\text{-}coast(s)$  is the distance of the location  $s$  to the nearest shore.

We thus assumed that the preference for fishing locations is a function of spatial location that differs intra-annually (time is measured as the day of year of a haul), depth and distance to the nearest shore.

Under the GAM framework,  $V(s, t)$  is defined as a full tensor product smooth between a spatial field  $s(x, y)$ , with  $x$  (longitude) and  $y$  (latitude) coordinates, defined as a Gaussian process smooth with a Matérn covariance function (Grüss et al., 2017; Stock et al., 2020; Maunder et al., 2020), and a temporal component  $f(t)$  defined as a cyclic cubic smooth. This 2-dimensional spatial smooth is able to account for any spatial autocorrelation apparent without demanding more technical “second-order” specification of autocorrelation (Hefley et al., 2017 as e. g. described in Dormann et al., 2007).  $depth(s)$  and  $distance\text{-}to\text{-}coast(s)$  were modeled using cubic regression smooths.

**2.2.1.2. Model evaluation and selection.** Apart from season and location, depth and travel time to the fishing ground constitute covariates that are influential on fisher decisions regarding gear deployment. Travel time is proportional to the distance to the nearest coast. Hence, in building the weighted point process model to capture the fishing intensity we tested all the simpler (nested) models than the one described in Eq. 1 (see Suppl. Table S1 and Suppl. Figure S2 for the model specifications) and compared the AIC values of the candidate models as well as their ability to correctly estimate reduced fishing intensities in areas further away from observed locations. The model presented in Eq. 1 presented consistently low AIC values and a larger relative drop in estimated intensity in un-fished areas compared to simpler models (as presented in

Suppl. Table S1, Suppl. Figure S2).

## 2.2.2. Catch standardization

**2.2.2.1. Hierarchical model.** We jointly modelled the catches of all species and fishing gears along a common spatio-temporal grid. Adding a spatio-temporal component in the model generally improves the standardization process and the performance of standardized catch as an index of abundance since spatial locations and temporal stamps are linked together through a proximity function spatially (covariance) or temporally (autocorrelation) and predictions are improved (see Zhou et al., 2019; Maunder et al., 2020, and references therein). To incorporate information available for all species we used varying smooths for each species for all covariates, i.e. “random slopes” in linear mixed model terminology, using a common penalty for the smooths, after accounting for inter-specific differences in their catches by using the species as a fixed effect in the models. This is an indirect way to jointly model all species rather than directly specifying a multivariate mixed model (Warton et al., 2015; Pedersen et al., 2019) but is a flexible approach under the GAM framework. To incorporate information among different fishing gears, we used: (i) the fishing intensity (Eq. 1) as a latent covariate in the catch standardization models and (ii) a random effect of selectivity calculated as an nested interaction between a fishing gear and its bin-transformed selectivity metric. Additional unexplained catchability effects were considered by including a random term for vessel (Rufener et al., 2021) as well as a smooth term for the vessel length.

To standardize catch by species  $i$  among all gears operating in each GSA 20 division we used the following GAM in simple notation (Eq. 2):

$$\text{catch}(i, s, t, v, g, y) = i + \text{loa}(v)|i + \text{depth}(s)|i + V(s, t)|i + \text{SST}(t, y)|i + \text{slope}(s)|i + \text{distance-to-coast}(s)|i + \log \lambda(s, t)|i + b_v(v)|i + b_g(g)|i + b_y(y)|i + b_{\text{sel}}(s, t, g, y)|i \quad (2)$$

where the catch of species  $i$  in location  $s$  and day  $t$  of year  $y$  caught by vessel  $v$  using gear  $g$  is modeled through a Tweedie family distribution (Hoyle et al., 2024) with variance estimated during fitting and a log link.  $i$  is a fixed effect for species.

$\text{loa}(v)$  is the overall length (as a proxy for the nominal effort) of vessel  $v$ .

$\text{depth}(s)$  is the depth at location  $s$ .

$V(s, t)$  is the tensor product smooth between the spatial Gaussian process  $s$  with longitude  $x$  and latitude  $y$ ,  $s(x, y)$ , and a cyclic cubic smooth of day of year  $t$ ,  $f(t)$ .

$\text{SST}(t, y)$  is the sea surface temperature at day  $t$  of year  $y$ .

$\text{slope}(s)$  is the bathymetric slope at location  $s$ .

$\text{distance-to-coast}(s)$  is the distance to the nearest coast of location  $s$ .

$\lambda(s, t)$  is the predicted -from the weighted point process model-sampling intensity at location  $s$  and day of year  $t$ .

$b_{\cdot}$  denote random effect “coefficients” (as random effect smooths under the GAM framework, set up using penalized splines) for vessel ( $v$ ), gear ( $g$ ), year ( $y$ ) and selectivity ( $\text{sel}$ ) deployed by gear  $g$  at location  $s$ , day of year  $t$  and year  $y$ . The  $b_{\cdot}$  effect was used as a proxy for any unexplained skipper effect. The remaining random effects ( $b_g$ ,  $b_y$  and  $b_{\text{sel}}$ ) are utilized in order to partially pool information across different years and gear selectivities since each year and selectivity type is neither independent of each other year and selectivity (absence of pooling and need to estimate separate effects) nor indistinguishable from each other year and selectivity (complete pooling under a single effect) but can rather be modeled as deviations from the model constant.

All non-parametric terms, except  $V(s, t)$ ,  $\text{depth}(s)$  and  $\text{distance-to-coast}(s)$ , were defined as cubic regression smooths.  $V(s, t)$  was defined as Gaussian process smooth with a Matérn covariance function while  $\text{depth}(s)$  and  $\text{distance-to-coast}(s)$  were defined as adaptive smooths. We opted for the use of adaptive smooths for these two covariates instead of transforming them in order to retain their original scales while also having a better predictive performance on the smaller values of each covariate. The  $\langle \text{term} \rangle | i$  notation denotes a smooth interaction between a  $\langle \text{term} \rangle$  and each species  $i$ .

**2.2.2.2. Model evaluation and selection.** Our main focus was to examine the causal effect of a selected set of covariates known to influence the catch of a fishing operation rather than to build the most parsimonious standardization model. Thus, by assuming that the selected covariates are relevant, the model predictive accuracy is not compromised (Kovács, 2024). For this, a full set of covariates commonly recorded on monitoring surveys or easily retrieved online was retained for the model (Stock et al., 2020; Hoyle et al., 2024). We applied the “double penalty” approach as described in Marra and Wood (2011) where the common penalty applied automatically to the non-linear basis functions during model fitting is enhanced with a second penalty applied to the linear basis functions of the covariates. This way, the effect of a term can potentially be canceled out during prediction (although not removed from the model) if deemed not relevant. The “double penalty” approach can be easily set during model fitting in the “mgcv” package without any manual tuning. Besides from the penalties applied to basis functions that are a part of the model fitting process, we examined, through trial and error, the basis type and dimensions available in the “mgcv” package to avoid over-fitting or over-smoothing of the terms. All preliminary basis type and dimension tests were performed on subsets of the full dataset

due to the large computational time needed to evaluate the full-dataset model. These subsets of the full dataset were sampled in a way that all fishing gears were retained while fewer species were included in order to reduce the number of species-specific effects that needed to be estimated (and hence computational time) during model fitting. Model summaries are presented in Suppl. Figure S3. For evaluation purposes, we examined the fit of the Eq. 2 models by checking for any patterns of the deviance residuals against all the covariates (Suppl. Figures S4a, S4b, S4c) as well as examining simulation-based residuals of the models (Suppl. Figures S5a, S5b, S5c). Finally, we inspected the existence of potential concavity among the smoothers of the partial effects that could affect their interpretation (Suppl. Figure S6), where a rough cut-off value of 0.5 has been shown as tolerable (Ramsay et al., 2003a; Ramsay et al., 2003b).

### 2.2.3. Evaluation of integrating multiple fishing gears

The use of fishing intensity in our catch standardization model apart from accounting for the preferential sampling, it also makes fishing gear integration possible. The alternative to integration is to carry out separate models for each fishing gear and then apply multi-model inference techniques to infer about the coefficients/smoothers and make predictions. However, averaging inference across models especially when fitted on different datasets (as in the case of the different fishing gears) is not a trivial procedure (Burnham et al., 2002; Grueber et al., 2011).

In order to quantify the improvement in the abundance estimates of our framework, we carried out separate catch standardization models for each gear in each GSA 20 division. The gear-specific model formulations were identical to Eq. 2 apart from the fact that the fishing intensity covariate was replaced with the effective fishing effort of each gear (length of nets, number of hooks and trawl duration). For each model we predicted inter-annual and spatio-temporal abundances for all species and their respective uncertainties. We, also, computed a simple “weighted average” estimate across all gear-specific models where the estimate of each gear-specific model was weighted with the amount of the data available for that gear. Because all models inherently included multiple species, we standardized the uncertainties against the species using a mixed model with a random effect for species.

### 2.2.4. Model applications

The standardized catches can be further analyzed for incorporation in applications in a more concrete way, e.g.: 1) obtain yearly abundance indices, 2) species distribution modelling, 3) determination of multiple-species hotspots in space and time, 4) identifying common response patterns to environmental/fisheries stressors across species and 5) investigating affinities between gears. These can be useful for species monitoring, informing stock assessments or designing mixed fisheries management measures (e.g. spatial/temporal closures, gear restrictions). Specifically: 1) Abundance indices that reflect the trend of a stock throughout the years can be estimated by the partial effect of year. 2) Using the spatial and temporal partial effects (coordinates, bathymetry, bottom slope, distance to nearest coast, SST and day-of-year) of the predicted catches we estimated the species distribution patterns. The predicted distributional patterns were evaluated on a spatio-temporal grid of resolution  $1 \text{ km} \times 1 \text{ km} \times 10 \text{ days}$  for each GSA 20 division. 3) We used the distributional patterns of multiple species to highlight areas and times of high abundance (hotspots) persistent across species. For

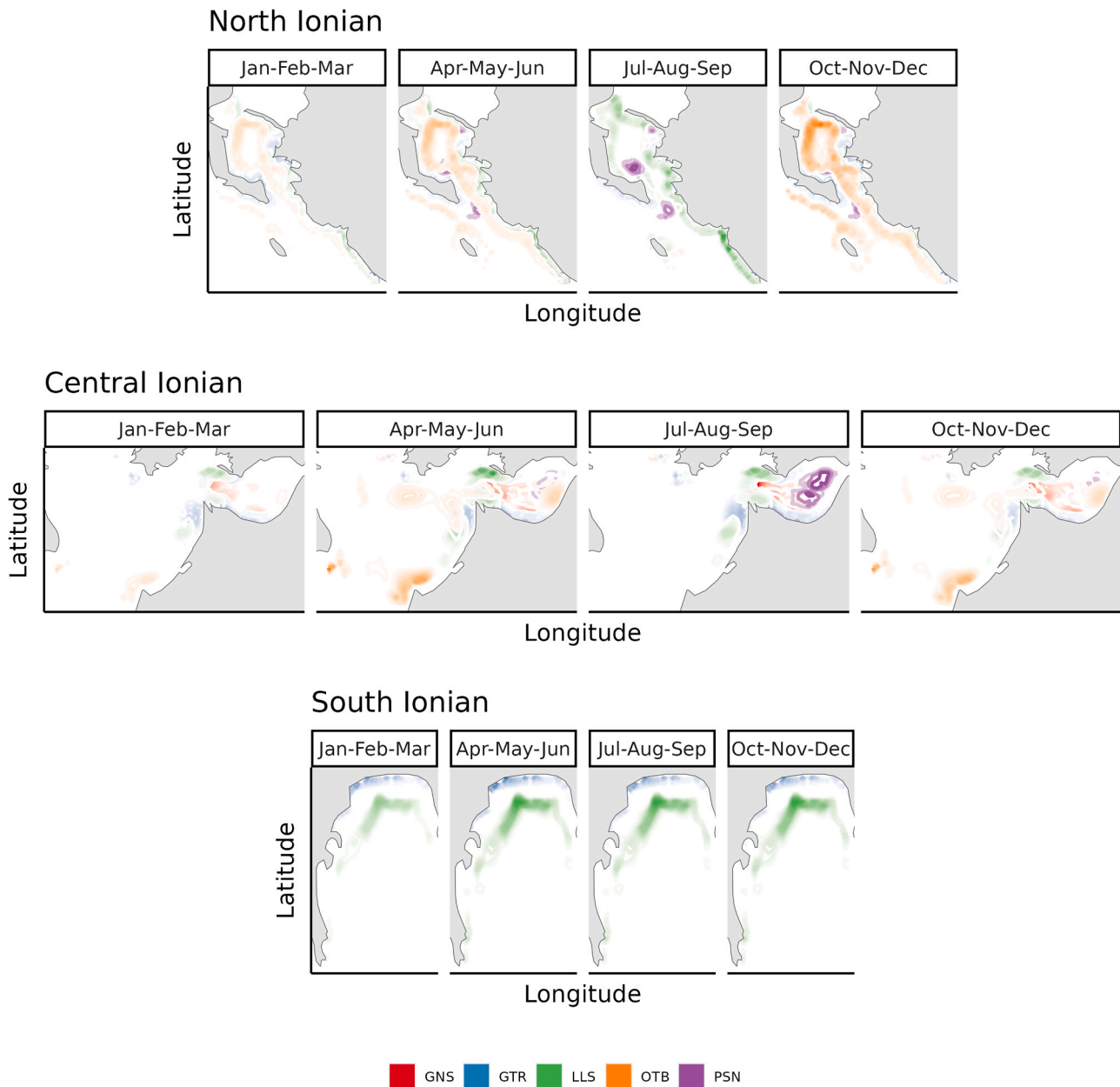
this we applied emerging hotspot analysis (Parry and Locke, 2024) using the Getis-Ord  $G_i^*$  (Getis and Ord, 1992; Ord and Getis, 1995) and followed the approach as detailed in Alglave et al. (2023). In short, using the standardized catches of each species in each grid cell, we computed  $G_i^*$  which assesses each cell alongside its neighboring ones to determine localities of cells with systematically low or high values. We then aggregated the cells classified as hotspots across all species to compute a persistence index of high standardized catches. 4) To determine common patterns among species we used k-means clustering to group individual species-specific catch responses. The appropriate number of groups was chosen by minimizing the total within-cluster sum of squares; the smallest number of groups that led to significant reduction of the sum of squares (besides which a threshold was apparent visually) was retained. 5) To examine fishing gear assemblages, we performed a multi-dimensional scaling (MDS) on the standardized species-specific catches across the fishing gear (fishing gear, gear selectivity and

skipper effect).

### 3. Results

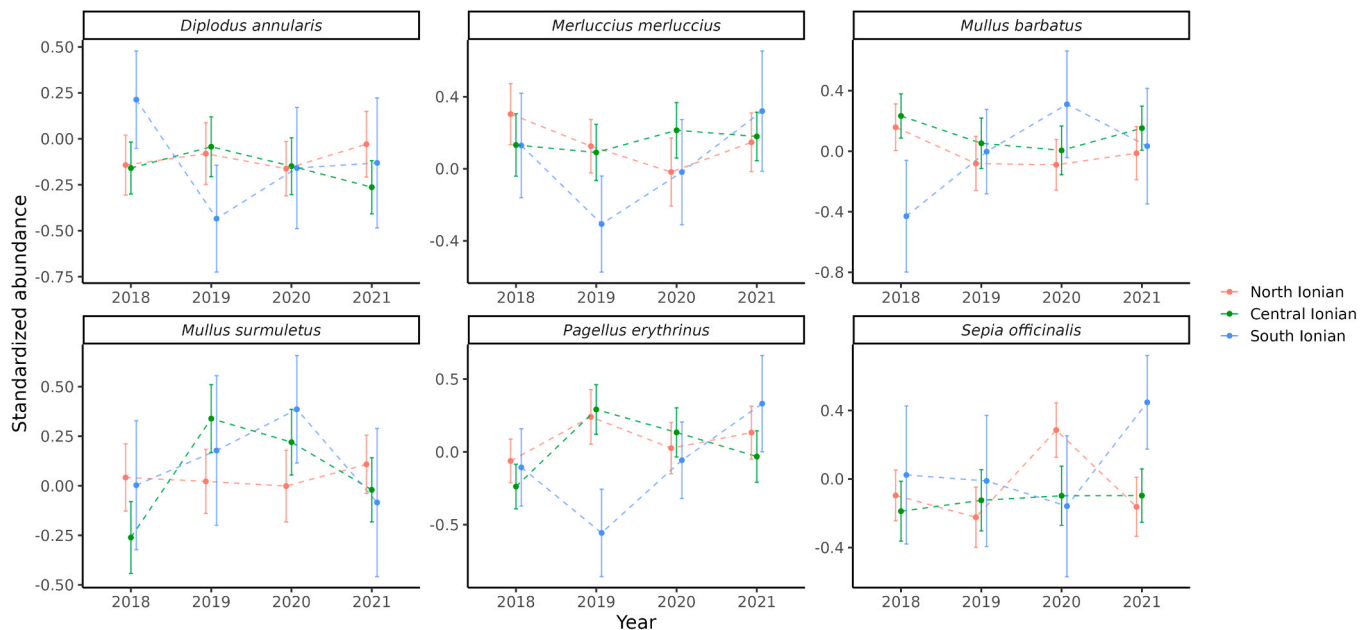
#### 3.1. Fishing intensity

Estimates of fishing intensity from the weighted point process model for each GSA 20 subdivision are shown in Fig. 3. The figure shows the predicted fishing intensity for each subarea and each fishing gear along a common spatio-temporal grid. The sampling intensity expresses the effective fishing effort as well as the spatial and temporal fisher bias. Especially for the Central and North Ionian regions, common fishing intensity patterns can be observed respectively for each gear. Trammel nets (GTR) operated near the coasts, at shallower depths, with higher intensity during the late spring to early autumn period. Purse seines (PSN) showed higher intensity during the summer period inside the



**Fig. 3.** Spatial and temporal allocation of the fishing intensity, for each fishing gear, as predicted from the generalized additive weighted point process model (Eq. 1). The intensity of the contour lines corresponds to the fishing intensity. Fishing intensity denotes sampling occurrence weighted by the effective fishing effort. The unit of fishing intensity is the rate of sampling occurrence and the scale differs among the three GSA 20 divisions; hence, no scale is depicted.





**Fig. 4.** Inter-annual trends of standardized catches of six species common in all GSA 20 divisions. Estimated abundance indices along with their standard errors are presented.

main gulf of each region at deeper waters. Bottom trawlers (OTB) had higher intensity before and after the summer trawl ban (as a result of the Greek legislation implemented in GSA 20). Gillnets (GSN) showed near constant intensities all year round at a variety of depths. Bottom longlines (LLS) showed higher intensities during the summer period at deeper/offshore fishing grounds.

### 3.2. Single-species assessment

Fig. 4 shows the estimated inter-annual variability of the standardized catch of six species present in all GSA 20 divisions. Larger fluctuations, as well as larger errors in the standardized catches, are present in the South Ionian which, compared to the North and Central Ionian, had less data (fewer observations) and also lower number of fishing gears (only trammel nets and set longlines). No clear temporal trends are evident in the abundance fluctuations (due to the small number of years and the error magnitudes), still there is indication of some increase in the standardized abundance of *Merluccius merluccius*, *Pagellus erythrinus*, *Sepia officinalis* in the last year or couple of years in that area.

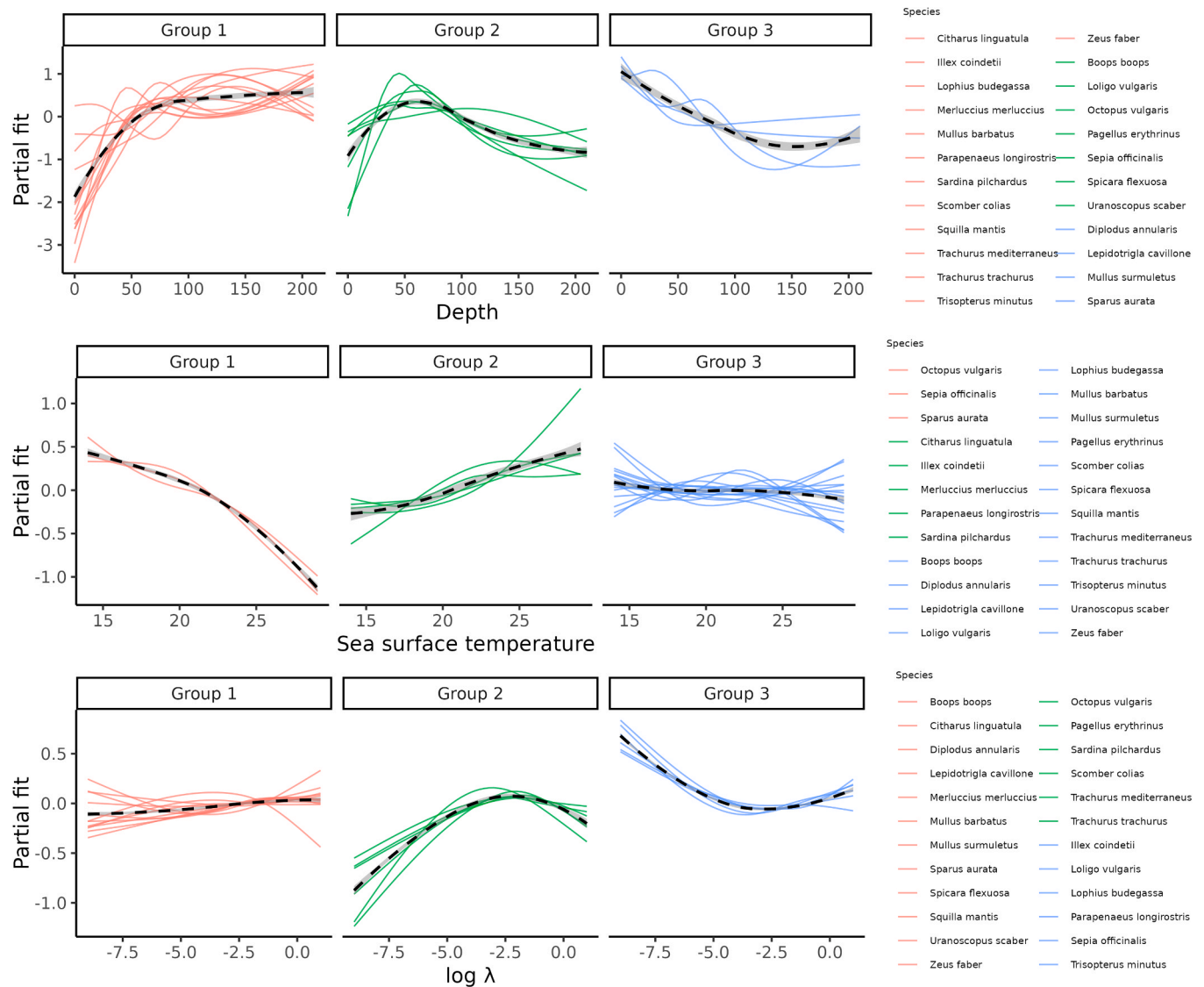
In addition to yearly abundance indices for specific species, the marginal effects of covariates, other than the year, on the species can be examined. Suppl. Figure S7a presents the univariate effects, after controlling for the rest of all other covariates, on the catch of *Merluccius merluccius* in the Central Ionian of GSA 20, as an example. The environmental covariates were incorporated in spatio-temporal predictions of abundance since they are linked to a specific space and/or time, while other covariates control for catchability and selectivity variation, i.e. the effect of the technical characteristics of the fishing gear as well as the unexplained skipper effect. In this example, we found specific gear types (e.g. longlines with large-sized hooks) that yielded larger catches compared to others. Additionally, we noted both inter- and intra-vessel variability in the catch of this species. Spatial and temporal distributions of the species for each GSA 20 division, can be retrieved from the standardized catches after catchability and other biotic covariates effects have been controlled for. Suppl. Figure S7b shows the spatial and temporal distribution of standardized catch for the *Merluccius merluccius* species in the Central Ionian of GSA 20 (accompanying uncertainty is presented in Suppl. Figure S7c). Throughout the year, we observe systematically higher abundances in deeper waters further from the shore.

An emerging pattern of higher abundance at the start of autumn is also evident as well as the migration of this pattern towards the closed gulf of Central Ionian towards the end of autumn.

### 3.3. Species assemblages

The segregated effects of selected covariates on the species of mixed fisheries catches is potentially useful in revealing inter-specific trends. The species-specific catch patterns, in each GSA 20 division, have been aggregated using k-means clustering (results of the analysis regarding the selected number of clusters are presented in Suppl. Fig. S8a and S8b). Fig. 5 shows the partial effect of depth, SST and fishing intensity on the species catch, respectively (specifically for the Central Ionian division). Similar patterns to the ones presented in Fig. 5 are observed for both the other divisions of GSA 20 (nonetheless, with different inter-specific trends) as presented in Suppl. Figs. S9a (North Ionian) and S9b (South Ionian). Regarding depth, we can visually separate the species into three distinct groups based on their estimated catch; species whose catch i) increases with depth, ii) increases with depth up to a point and then declines, and iii) declines with depth. The species estimated catch in response to sea surface temperature has, also, been discretized in three groups: i) catch declining with SST, ii) catch increasing with SST, and iii) catch mostly unaffected by SST. The effect of fishing intensity on the species catch can be separated in three groups, too: i) species whose catch more or less increases with fishing effort and species that are more sensitive to effort fluctuations and display either ii) an increase in catch with small effort increases and then a catch decline with larger effort increases and iii) a decline in catch with small effort increases followed by increased catch with larger effort increases. The key aspect in the two latter groups is the catch response in higher effort, since this is of ecological value, which, essentially, reduces catch response to groups of species whose catch either decreases or increases, respectively, as sampling intensity increases.

Fig. 6 presents areas of high estimated abundances (as predicted from the hotspot analysis) that persist across species. We can see that a temporal trend is evident since, throughout the year, the areas of high abundances either move spatially (for example, high abundances appear inside the gulf of the Central Ionian towards the end of the year) or change in intensity of persistence (for example the number of species



**Fig. 5.** Estimated abundance responses of the most common commercial species of the Central Ionian (GSA 20) relative to depth, SST and fishing intensity ( $\lambda$ ; on log scale). The partial effect of each covariate is shown (with all other covariate effects canceled out). To aid visualization, similar inter-specific patterns have been grouped using k-means clustering. The dashed line represents the trend of each cluster.

with high estimated abundances increases through the year in the gulf of North and South Ionian).

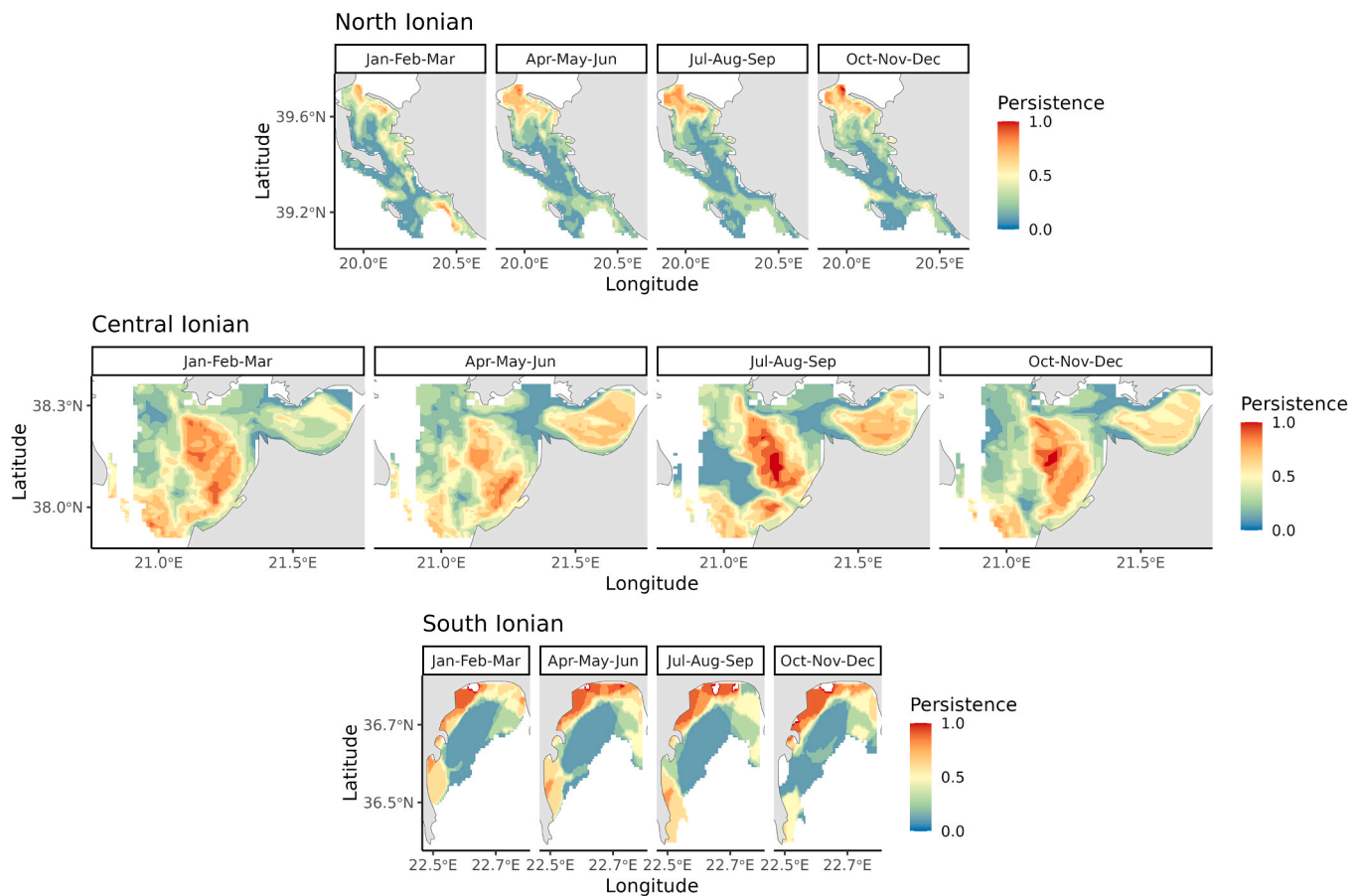
### 3.4. Fishing gear assemblages

Fig. 7 shows the MDS of the different gear and selectivity combinations for all divisions of GSA 20. Each point represents a unique vessel which is attributed with either one or more fishing gears and selectivity bins (categories). We note a wide array of different emerging patterns within and between fishing gears in the divisions of GSA 20. In the North Ionian there is a clear aggregation of the distinct selectivity bins (20–26, 26–32, 32–36, 36–44) of trammel nets (GTR) which seem to have a distinct profile catch compared to other gears. The gillnets (GNS), on the other hand, show high heterogeneity as they show affinity with a variety of fishing gears and selectivities. In the Central Ionian, we observe an interesting pattern of a vessel displaying a unique profile catch irrespective of the different gears deployed. In addition, bottom trawlers (OTB) and purse seines (PSN) seem to possess a more similar profile of standardized catches when compared to small-scale fisheries gears. In the South Ionian, the deployed fishing gears are very few to provide sufficient context for large assemblages to appear, wherein the different

selectivities of longlines (LLS) are characterized with distinct profiles of catches.

### 3.5. Performance of the integrated model in reducing uncertainty of estimates

Fig. 8 presents the uncertainty of the yearly (A) and spatio-temporal (B) abundance indices as estimated from the multi-gear model described in Eq. 2 as well as the single-gear models. The uncertainties have been standardized across species in order to reveal the general trends. An important aspect of retrieving robust abundance indices to be used in assessment is for the uncertainty of the estimate to be i) low and ii) consistent across the covariate of interest. In the current case, the uncertainty of the inter-annual abundance indices is the lowest for the multi-gear model and also consistent across years for all GSA 20 divisions (Fig. 8A). Additionally, the estimated uncertainty extends to lower ranges both in the spatial and temporal (intra-annual) grid compared to single-gear models in a more systematic manner (Fig. 8B). Since, for visualization purposes, Fig. 8B presents the summarized uncertainties along space and time, finer-scale uncertainty estimates are presented in Suppl. Figures S10a and S10b.



**Fig. 6.** Spatio-temporal distribution of the persistency of high estimated abundances across species in the GSA 20 divisions. The persistence scale ranges from 0 (no species has high estimated abundance) to 1 (all species have high estimated abundances).

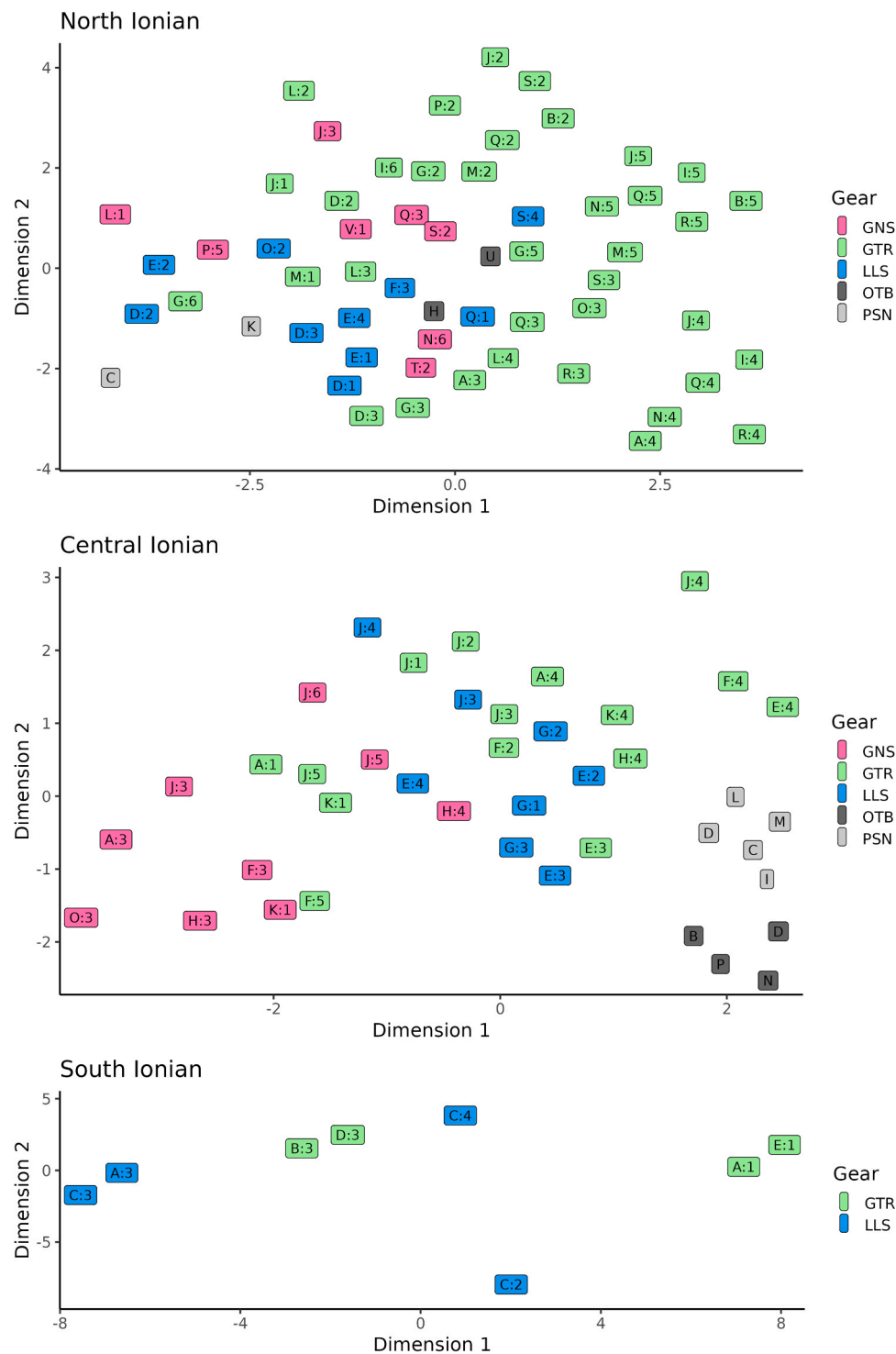
#### 4. Discussion

In the current study we examined aspects of typical multi-gear mixed fisheries using the GSA 20 (eastern Ionian Sea) as a case study. Our focus was on standardizing the catches of selected commercial and/or abundant species representative of the catch profile in this specific area. This information can be useful for planning and improving the management of mixed fisheries both on the scale of stocks (species) and on that of gears (whose technical characteristics or spatiotemporal restrictions are often the focus of management), but may also be useful for stock assessments, the comprehension of population ecology or the study of fisher behavior and gear-stock interactions.

The data used in the current study constitute a typical example of sources of information that are usually available from fisheries monitoring schemes (e.g. the EU Data Collection Framework). Contrary to experimental/survey data that follow standardized protocols (e.g. Bertrand et al., 2002), allowing for the collection of fisheries-independent data usually from a network of sampling stations on a specific time of the year, fisheries monitoring data contain a vast amount of information regarding spatial and temporal coverage. However, a major disadvantage of commercial fisheries data is the inherent bias during the sampling procedure (Bishop, 2006; Cotter and Pilling, 2007; Maunder et al., 2020; Howard et al., 2023). In addition to possible biases introduced due to monitoring sampling protocols and the deployment of observers, another main source of bias is traced to fisher behavior and can conceptually be thinned out to four main components: i) the spatial and temporal allocation, ii) the type (i.e. the unit of reference), iii) the quantity (i.e. the amount) and iv) the selectivity of the fishing/sampling effort. This, from an analysis perspective, translates to i) over- and under-fished (and thus sampled) areas and seasons, ii) a lack of a

uniform measure for the unit of sampling effort (e.g. rate of catch per fishing time/duration, per gear length etc.), iii) species-specific catch responses to effort (i.e. the catch of certain species does not relate linearly to the amount of allocated effort, while the catch of other species does) and iv) a lack of a catchability mapping among different technical characteristics of fishing gears (e.g. the way the catchability of small-sized hook longlines compare to bottom trawlers and whether there is a species-agnostic relationship between them).

In the current study we opted for the flexibility and ease of use of Generalized Additive Models to incorporate inter-gear and inter-specific variation in a joint analysis utilizing as much as possible shared information among fishing gears and species. For this we have used a two-step modeling approach where, firstly, the preferential sampling of each gear was accounted for (Eq. 1) and, subsequently, used as a control variable in a mixed effects joint model of species catches (Eq. 2), using a set of explanatory and control covariates allowing for inter-specific common and different patterns to be revealed. A more common recent practice, under the bayesian framework, is to model preferential sampling and abundance using a joint likelihood model with shared predictors (Pennino et al., 2019; Alglave et al., 2022). Our two-step GAM approach has the disadvantage of not robustly propagating variance to the standardization model (see, for example, Bravington et al., 2021, for an approach in Density Surface Models), while the bayesian framework presumably better accounts for that. Our attempt in eliminating fishing effort of commercial fisheries data was based on data as recorded by observer programs where a fishing operation is point-referenced (by the midpoint of start and end coordinates) and attributed a recorded fishing effort measure. More efficient proxies of fishing effort have been proposed (Mendo et al., 2023) which can be utilized when estimating fishing intensity through a point process with survey weights. In our

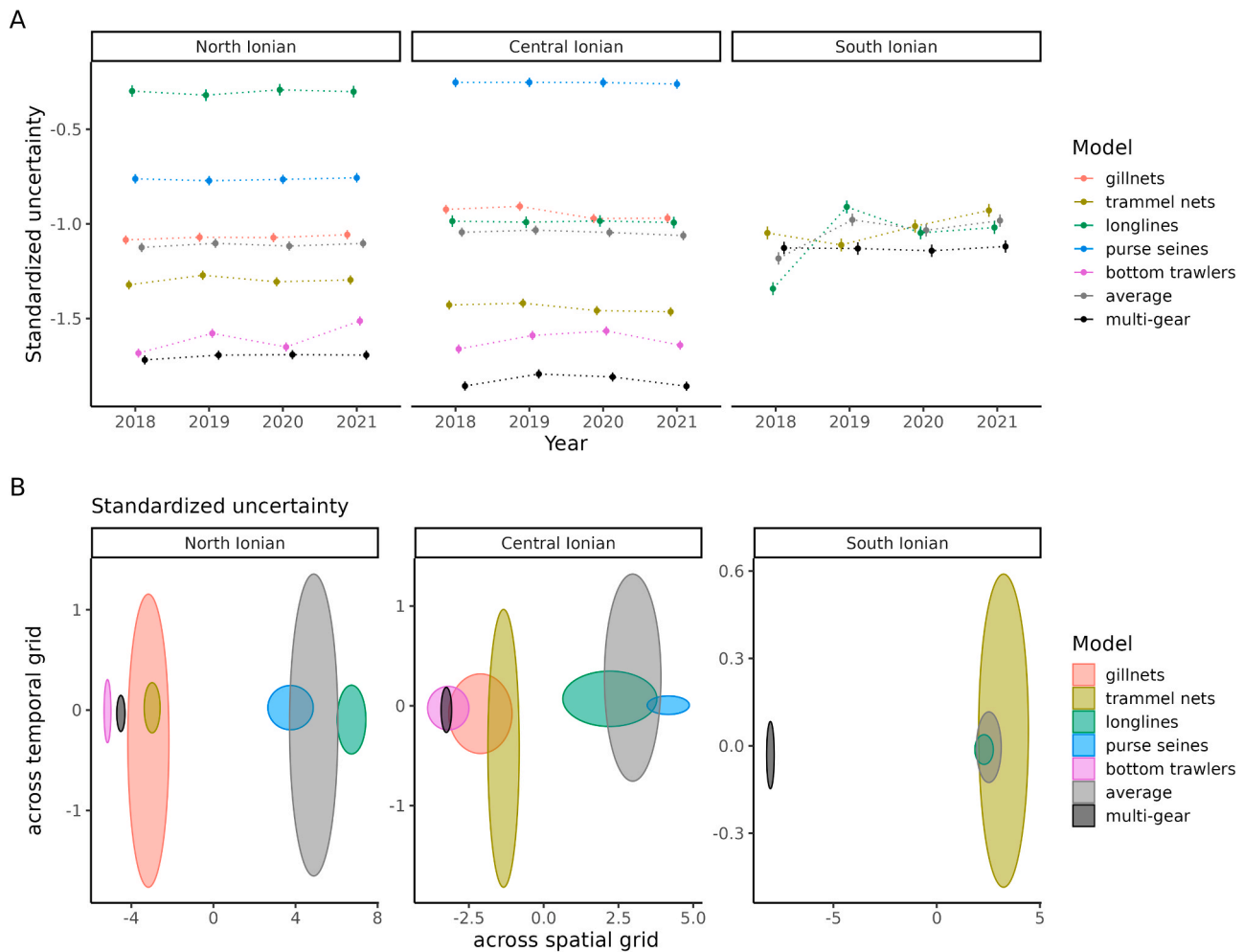


**Fig. 7.** Multi-dimensional scaling visualization of the standardized catch compositions of the fishing gear and selectivity combinations along with inter-vessel individual differences. The  $n$ -dimensional catch profile from the  $n$  species is reduced to 2 dimensions. Uppercase letters denote the distinct vessels. Numbers 1–6 denote the selectivity bins for each gear; for interpretation purposes, low numbers reflect gear selectivities that capture small individuals (small mesh-size, large hook-number) while high numbers reflect selectivities that capture large individuals (large mesh-size, small hook-number). Mesh sizes for gillnets (GNS) and trammel nets (GTR): 1:  $\leq 20$ , 2: 20–26, 3: 26–32, 4: 32–36, 5: 36–44, 6:  $> 44$ . Hook numbers for longlines (LLS): 1:  $> 14$ , 2: 11–14, 3: 8–11, 4:  $\leq 8$ .

work we used a single model to integrate data from all fishing gears. Alternatives in integrating different datasets that share common response and covariate variables exist under the bayesian framework using joint likelihood (Fletcher et al., 2019; Paradinas et al., 2023) which potentially requires adjusting the datasets selecting differing weights (Punt, 2017). By jointly modeling many species using

species-specific smoothers we can share inter-specific information of distribution which can be beneficial in assessing rare species (Stock et al., 2020). Various methods have also been proposed to model multispecies responses that accommodate more sophisticated residual covariance among species (Thorson, 2019, using VAST; Roberts et al., 2022, using bayesian joint modeling). Nonetheless, we note that the





**Fig. 8.** Uncertainty of predicted abundance indices on an annual (A) and spatio-temporal (B) level. Uncertainties are retrieved from the i) single-gear models, ii) averaged multi-model estimation across the single-gear models and iii) multi-gear model of Eq. 2. The uncertainties of the estimates have been standardized against species to remove any species-specific effects that influences uncertainty. Panel A shows the fluctuation of uncertainty throughout the years of the study. Panel B shows the range that uncertainty spans across the spatial grid (x axis) and the temporal grid (y axis) of the model spatio-temporal field. Uncertainty in Panel B has been summarized using median bounded by the first and third quartile for space and time, respectively, for visualization purposes. Supplemental Figs. S10a and S10b present the fine-scale uncertainty along space and time.

main strengths of our proposed GAM framework is flexibility, wide applicability and ease-of-use. We believe our modeling approach can sufficiently extract the complexities that accompany the multi-gear multi-species fisheries and estimate stock abundance indices in a way that can be applicable in a management context.

Sampling intensity (here represented as fishing intensity) is commonly modeled through the use of point process models. Especially, when using georeferenced data, the class of log-Gaussian Cox processes, enhanced with the efficiency of integrated nested Laplace approximation (INLA) and stochastic partial differential equation (SPDE) models is the main approach adopted in analyzing spatial point processes (e.g. Pennino et al., 2019; Watson, 2021). In the current work we opted for a point process GAM where, as explained by Youngman and Economou (2017), the intensity of a point process is modeled by decomposing a set of explanatory covariates into basis functions and inference is based using restricted likelihood, which i) offers flexibility and ii) eliminates the need to aggregate occurrence data on an arbitrarily defined grid in contrast to the most common methods. Using this GAM framework we modeled fishing intensity as a function of covariates that affect fisher behavior: i) the spatial and temporal stamp of a fishing operation, ii) the depth and iii) the distance from the coast needed to be traveled for the fishing operation to take place. The topography of our study area (eastern Ionian Sea) is characterized as a steep coastline that may reach

large depths relatively close to the shore, while, on the other hand, there are a number of gulfs (e.g. Patraikos Gulf in the Central Ionian, Kerkiraikos Gulf in the North Ionian) where this pattern is not followed and, hence, both bathymetry and distance to coast equally influence fisher decisions. Additionally, distance to coast is a variable used commonly to define fishing regions under national enforced management schemes and, thus, is of specific importance. Of course, other economic, cultural and managerial factors also influence fisher behavior, and hence the distribution of fishing intensity (Wilén, 2004; Oostdijk et al., 2024), but our focus here was to utilize covariates that are readily collected/available in typical fisheries monitoring schemes. This full set of covariates was retained in the point process model since it displayed lower AIC values and better estimating ability compared to simpler point process models (Suppl. Table S1, Suppl. Figure S2). Besides the qualitative aspect of fisher bias, we aimed to incorporate the quantitative aspect as well, i.e., the effective fishing effort that is an attribute in each occurrence of a sample/fishing operation. For this, we inserted the effective fishing effort as weights in the point process models so that each fishing operation contributed to the overall sampling intensity relatively to the fishing effort. Since the effective fishing effort was incorporated in the point process models as weights, the need to control for the unit of effort of each fishing gear in the catch standardization model was eliminated. Thus, each separate point process

model, for each fishing gear, has “absorbed” the fishing effort accompanying each catch value and, subsequently, the fishing intensity, which is expressed as a rate of sampling occurrence (in log scale), can be used as a control covariate in the catch standardization model as directly comparable among gears (Fletcher et al., 2019). As such, we can combine data from different gears and, as we have shown above, this integration decreases the uncertainty when extracting yearly abundance indices compared to single-gear models. This is true not only for the yearly trends but for all other effects presented here. Integration of fishing gears increases the spatial and temporal coverage and hence the quality of the estimations (the quality of the estimations refers to both how inter-gear conflicting trends are balanced and to the reduced uncertainty of the estimations). Integration also pools species that otherwise constitute targets of different gears under a common framework and hence the multi-species aspect of mixed fisheries can be examined.

The allocation of fishing intensity for each gear and each GSA 20 subdivision shows inter-gear and inter-division variability. Target species ecology as well as local management measures and level of implementation drive the fishing effort patterns exhibited from all fishing gears operating in GSA 20. Bottom trawlers operate mostly before and after the summer ban that is applied on this gear. Also, especially in the Central Ionian, the spatial coverage of the bottom trawlers is much different from the other gears operating on this division since trawlers are mostly operating in the open sea rather than inside the central bay formed in the area (Patraikos Gulf, where bottom trawls are only allowed to operate from December to February). Purse seines show a peak in their activity during spring/summer where weather conditions are favorable for this gear (banned in January and February). Fishing gears that belong to the small-scale fisheries (set longlines, gillnets and trammel nets) have a more consistent annual coverage since no management rules dictate seasonal bans of small-scale fisheries in GSA 20. While the temporal patterns of fishing intensity in industrial gears (trawl, purse seine) are dictated by legislation (seasonal gear prohibitions), the seasonal and spatial dynamics in small-scale fishing gears can be useful to plan gear-specific management measures in space and time. As an example, in cases where high fishing intensity coincides with low catches, measures can be taken to reduce the fishing effort exerted without significant losses in fisher profit. Such example cases present the longlines in North and South Ionian and trammel nets in the Central Ionian, where effort is high (Fig. 3) while catch is low (Fig. 6). From the above, it is evident that a standardization of commercial catch data to use as proxies for stock abundance and distribution using only one fishing gear can lead to different estimates than using catch data from another gear and that common standardization techniques among gears are essential (Gibson-Reinemer et al., 2017).

Designing and establishing management regimes for mixed fisheries that can furthermore be effective for the variety of stocks targeted by the multiple gears is a tedious task. It is advised in multispecies catch standardization techniques to account for the fishing tactics (métiers) employed by fishers when targeting species assemblages (Winker et al., 2013; Thorson et al., 2017). In the current work we adjusted for fishing behavior through controlling for fishing effort and selectivity using species-specific smoothers under the GAM framework. The selectivity implemented by each gear, as defined by its technical characteristics, which in the model described here is incorporated as a very rugged adjustment for the fishing tactics observed, implies that the fishing tactic can act as a reference managerial unit when referring to fishing gears (Tzanatos et al., 2013). Each division of GSA 20 has its own inherent unique set of common species appearing in the catch (Table 2) which along with the variability in inter-specific patterns of catch along space and time and the inter-gear catchability patterns pose difficulties in proposing a single managerial scheme for GSA 20 as a unit. Finer scale management schemes, on more local scales, seem to be better suited in accommodating area-specific fishing practices (Martin Gonzalez et al., 2021) as our results have, also, suggested since many trends with regards to species and gear assemblages are contrasting among the GSA

20 divisions.

Advancing from model outputs to management advice that is actually applicable is a cumbersome step (Karp et al., 2023). As presented in our work, in the context of mixed fisheries there is a multitude of information that needs to be interpreted. Here we thinned out the information to three inference aspects relevant to management: (i) that concerning a specific stock, (ii) shared patterns among species and (iii) shared patterns among fishing gears. Indices of abundance as developed in the current work (Fig. 4) are essential to stock assessment (Campbell, 2015). Although in our case, the annual coverage of catches was not large to observe large-scale inter-annual trends (our data were collected in the period 2018–2021), a dataset that includes larger time-series could illustrate clearer trends. It is worth noting, also, the importance of integrating multiple fishing gears into the standardization procedure which is evident from the case of the South Ionian. In the South Ionian, due to the observer monitoring program sampling only two fishing gears, the uncertainty in the estimation is larger compared to North and Central divisions of GSA 20 as well as there seems to be an instability of the model to output a trend. Besides the yearly abundance indices, in the specific examples presented from GSA 20, it is evident that a single species can display different spatial distributions in each division as determined by the local collective fishing behavior (allocation of fishing effort, deployment of fishing gears, heterogeneity between fishing trips of one fisher or between fishers-skipper effect). From a management perspective, accounting for spatial and temporal abundance variations seems straightforward, but not adjusting for fisher variability can lead to unexpected outcomes since a static management scheme is prone to changes of fishing behavior (Salas and Gaertner, 2004; Oostdijk et al., 2024). Indeed, our example shows that the skipper effect is significant (see the partial effect of vessel ID on Suppl. Figure S7a, as well as Fig. 7). Within a certain métier (in the current work simplified to the level of a gear's selectivity), certain skippers illustrate high variability in their daily catches while others are more consistent in their effort and catches. Also, some métiers illustrate low heterogeneity among vessels (for example trammel nets of small mesh size in the North Ionian, while others illustrate high heterogeneity (for example gillnets in the Central Ionian).

Regarding the improvement of observer/monitoring schemes, the standardization results could help redirect sampling effort towards gears/métiers with high intrinsic variability (complementary to the other criteria used to allocate sampling effort). For example, re-allocating observer resources from industrial gears towards small-scale fisheries gears can prove to be a valuable investment in the Central Ionian (Fig. 7). On the other hand, allocating observer resources, in a stratified manner, towards more systematic monitoring of fishers deploying distinct trammel nets selectivities in the North Ionian and distinct longlines selectivities in the South Ionian is a rational step considering the low variability evident within each selectivity compared to the high variability between the distinct selectivities (Fig. 7).

Management could utilize the findings in shaping restrictions that are both biologically (using biological reference points) and financially (providing some minimum catch and resulting profits) sustainable. Furthermore, sharing catch information among species can be valuable in changing the reference unit from a stock to an assemblage of stocks whose catches display common patterns across environmental variables or fishing effort. This can enhance management measures to include more species whose estimated abundances are affected similarly. In the examples presented here, some clear grouped species responses were formed relative to specific effects (Fig. 5 and Suppl. Figure S9). This grouping of species could allow the design of common management measures, especially for species that are caught by one or more common gears. The inclusion of longer time series in the standardization process would naturally allow for the determination of clear trends in the dynamics of stock abundance and in the identification of common or contrasting patterns across areas that might be used in management. The findings concerning the spatio-temporal distribution of species can be

used to design spatial restrictions, either in the context of single-species management (e.g. if management authorities would opt for the improvement in the yield of select specific stocks) or under a multi-species context (that might incorporate information from the “community” of the species examined) that could define multi-stock persistence maps of abundance (Fig. 6). On the same note, assemblages of fishing tactics that affect stocks in a similar manner, rather than a unique fishing gear, can comprise a reference unit for management (Fig. 7). In the examples here, we observed an aggregation of different fishing gears that target large individuals (e.g. gillnets and trammel nets of large mesh size and long lines with large-sized hook) while the same gears targeting smaller individuals (small mesh size and small-sized hook) show a more fragmented catch composition. This size-based selectivity of fishing gears can provide a seamless extension of our current work by standardizing the catches at age for each species, rather than total catches. Length data are commonly recorded during observer monitoring programs which along with complementary information (from literature) of species maturity lengths can help in segregating the mature and immature portion of the catch for each species. This can be useful in identifying marine areas of potential interest like spawning and nursery grounds (see, e.g. [Alglave et al., 2023](#)). Another extension of our work, under the management perspective, would be the inclusion of covariates endowing societal and economic factors that affect both the allocation of fishing effort and the observed catches. Such covariates can include more fine short-term fishing strategies (rather than selectivity of gears) and information on the potential earnings and expenses of a fishing and their fluctuations with regards to current market, albeit such information is not readily available or collected at a large spatial and temporal scale similar to catch data records.

Catch standardization is an essential tool for stock assessments ([Maunder and Punt, 2004](#); [Campbell, 2015](#), [Hoyle et al., 2024](#)). Here, we have addressed challenges faced in modern standardization approaches like accounting for spatial and temporal processes ([Thorson, 2019](#); [Maunder et al., 2020](#)), removing the potential biases introduced due to fisher preferential sampling ([Pennino et al., 2019](#); [Alglave et al., 2022](#)) and joint modelling of species ([Warton et al., 2015](#); [Thorson et al., 2017](#); [Roberts et al., 2022](#)). Furthermore, we have turned the significant inference obstacle of the existence of multiple gears and heterogeneous métiers typically present in mixed fisheries ([Duarte et al., 2009](#), [Zhou et al., 2014](#)) into an asset by integrating data from a variety of gears under a common framework. This approach can extend the commercial catch standardizations typically performed at gear level (depending on the most representative gear sampling a target species; e.g. [Alglave et al., 2022](#), utilizing bottom trawlers in their examples; [Paradinas et al., 2023](#), utilizing trammel nets in their examples) to multiple gears and be applied to a considerable number of “fisheries systems” worldwide that are typically multi-gear and multi-species in order to provide objective estimates of (unit or multiple) stock abundance. Subsequent steps, like the example applications and extensions of the standardization process we have provided in our work, can help set up a useful toolbox to inform fisheries scientists and managers for the improvement of monitoring/sampling schemes or designing measures at the spatial, temporal or gear-métier level.

#### CRedit authorship contribution statement

**Kavadas Stefanos:** Writing – review & editing, Data curation. **Tzannatos Evangelos:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Lazaris Alexis:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Tserpes George:** Writing – review & editing, Methodology, Conceptualization.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.fishres.2025.107305](https://doi.org/10.1016/j.fishres.2025.107305).

#### Data availability

The authors do not have permission to share data.

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